

# Subham Karmakar

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## Education

Adamas University- B.Tech, Computer Science and Engineering | CGPA: 9.0/10

Aug 2024 – Jul 2028, Kolkata, India

## Technical Skills

Languages: TypeScript, JavaScript, Python, Go, C++, SQL

Frontend: React, Next.js, Tailwind CSS, Electron, React Native(Expo Go), HTML/CSS3

Backend: Node.js, Express, FastAPI, REST APIs, WebSockets, Server-Sent Events

Tools & Infrastructure: Git, Docker, Docker Compose, GitHub Actions, Nginx, Linux

AI & Agentic Systems: LLM APIs (OpenAI, Anthropic, Gemini), OpenRouter, Ollama, LangChain, retrieval-augmented generation (RAG), conversational memory, semantic chunking, embeddings, vector search, structured AI actions, streaming responses.

Machine Learning: Python, scikit-learn, XGBoost, Random Forest, probability calibration, model evaluation.

Vector Storage & AI Backend: FastEmbed, sqlite-vec, pgvector, SQLite, PostgreSQL, FastAPI

## Experience

- **Ryze AI — Frontend Developer Intern** Jan 2026 – Apr 2026 | Remote
  - Shipped production UI components using React, TypeScript, and Tailwind CSS; participated in code reviews and sprint delivery.
  - Integrated REST APIs and implemented component-level caching to improve data-fetching behaviour.
- **Adamas University — Moodle LMS Cloud Maintainer, Part-time** Nov 2025 – May 2026 | Kolkata, India
  - Administered a cloud-hosted Moodle LMS supporting 1,000+ students and faculty, with responsibility for health monitoring and incident response.

## Selected Projects

1. **CYCODERS | Community Platform** - Next.js, TypeScript, Express.js, PostgreSQL, Docker, Nginx | [cycoders.in](https://cycoders.in)
    - Built a community platform serving **600+ active users**, with member dashboards, QR-code event check-in, and points-based progression and leaderboards.
    - Implemented **JWT authentication and role-based access for administrative and member workflows**, including event registration and attendance management.
  2. **Ruse | Temporary Screen-Sharing Platform** - Go, Next.js, TypeScript, Redis, LiveKit, WebSockets, Docker | [ruse.monostack.in](https://ruse.monostack.in)
    - Built a Go backend and Next.js interface for temporary collaboration rooms supporting screen sharing, camera, microphone, and ephemeral chat through LiveKit.
    - Implemented owner-approved admission, Redis-backed room state and capacity reservations, hashed session tokens, and **15-minute media access tokens** with publishing permissions.
  3. **SecureMailScope | Mail-Transport Security Assessment** - Python, FastAPI, XGBoost, Random Forest, scikit-learn, PostgreSQL, NextJS, TypeScript, TailwindCSS | [securemail0day.vercel.app](https://securemail0day.vercel.app), [github.com/Groy416/SecureMail-ML-Backend](https://github.com/Groy416/SecureMail-ML-Backend)
    - Contributed Dashboard, ML and backend components to an SIH team project that assesses **SMTP, IMAP, and POP3** session features using calibrated classifiers and evidence-linked security rules.
    - The project's XGBoost classifier achieved **98.00% accuracy and 0.9763 macro-F1 on 7,665 held-out synthetic packet-derived sessions**, with separate training and calibration partitions.
    - Contributed to a FastAPI backend supporting session analysis and packet-capture processing, combining XGBoost/Random Forest predictions with deterministic findings that remain authoritative in the final risk assessment. Along with that also developed the dashboard for giving users and clients a smooth experience.
- Atlas | AI Desktop Assistant** - Electron, TypeScript, Python, FastAPI, SQLite, sqlite-vec, FastEmbed | [github.com/Subham12R/Atlas](https://github.com/Subham12R/Atlas)
- Built a desktop assistant with a FastAPI backend and adapters for **OpenAI, Anthropic, Gemini, OpenRouter, and local models** through Ollama, supporting streamed responses and image inputs.
  - Implemented persistent conversational memory using **384-dimensional embeddings**, semantic chunking, and sqlite-vec search to retrieve relevant context from stored conversations.
  - Added rolling summaries and entity-relationship memory, with graceful fallback when summarisation is unavailable, alongside Server-Sent Events streaming and voice transcription.
5. **Noted | Collaborative Notes and Study Platform** - Next.js, TypeScript, Node.js, Yjs, Socket.io, PostgreSQL, Redis, Stripe | [notezero.vercel.app](https://notezero.vercel.app)
    - Built a collaborative note-taking application using **Yjs CRDTs and Socket.io** to merge document updates and share participant presence, with Redis-backed state snapshots.
    - Implemented AI-assisted actions for creating, updating, and organising notes, folders, and tasks, with validated action inputs and ownership checks.
    - Integrated Stripe Checkout, customer billing-portal access, and subscription webhooks to manage paid-plan lifecycle events.

## Leadership & Achievements

- Student Lead, CYCODERS: Community leadership following multiple Technical Lead roles for CYCODERS & ACM Student Chapter.

## Certifications & Training

- Oracle Agentic AI Certified Foundations — Aug 2026
- AWS Academy Graduate — Cloud Foundations and Cloud Architecting — Sep 2026
- Full Stack Development Foundations — Udemy — Jan 2025
- Claude Code 101 and MCP Introduction — Anthropic Education, Jun 2026